Zhi Chen

Website: zhichen3.github.io/
Email: zhi.chen.3@stonybrook.edu
ORCID: 0000-0002-2839-107X

Education

- **Stony Brook University** — Stony Brook, NY
  Doctor of Philosophy - Physics; — *Aug 2023 - Present*
- **Stony Brook University** — Stony Brook, NY
  Master of Arts - Physics; GPA: 3.94/4.00 — *Aug 2021 - May 2023*
  Thesis: *Sensitivity of He Flames in X-Ray Bursts to Nuclear Physics*
- **Stony Brook University** — Stony Brook, NY
  Bachelor of Science - Physics; GPA: 3.96/4.00; Summa Cum Laude — *Aug 2017 - May 2021*

Research Experience

- **PhD Thesis Research** — Aug 2023 - Present
  Graduate Student Research
  - Primary Project: Implemented axisymmetric spherical geometry in *Castro* for multidimensional simulations of helium deflagration in Type I X-ray bursts to study the influence of rotation on the burst dynamics.
  - Secondary Project: Implemented nuclear statistical equilibrium (NSE) evolution algorithms in *Microphysics* for efficient high-temperature reaction network integration in large-scale astrophysical simulations.
  - Software Development: Contributor in multiple astronomy software including *Pynucastro*, *Microphysics*, *Castro*, and *pyro2*.
- **Master Thesis Research** — Jan 2022 - Aug 2023
  Graduate Student Research
  - Primary Role: Investigated how different reaction networks, electron screening routines, and integration methods affect lateral Helium flame propagation in Type I X-ray bursts using *Castro*, an adaptive mesh, compressible hydrodynamics simulation code.
- **SULI Program at Brookhaven National Lab** — Jun 2021 - Jan 2022
  Summer Internship + Continued Research
  - General Setting: Worked in a small group developing a technique called two-photon interferometry for measuring the relative separation of two luminous objects with high precision.
  - Primary Role: Developed a *simulation code* that simulated the theoretical observational signal through two-photon interferometry given a set of telescopes and star pairs using Python. With the simulated signal, I used Markov Chain Monte Carlo algorithm to predict the precision of the relative separation between the star pair.
- **SULI Program at Brookhaven National Lab**
  Summer Internship — *Jun 2020 - Aug 2020*
  - General Setting: Worked in an Astro-group for building a radio-telescope to detect 21-cm emission from hydrogen
  - Primary Role: Developed additional features for *imcurio*, which is a simulation code for 21 cm intensity mapping observations that simulates a fixed set of visibilities for a fixed telescope and fixed sky using Python.

Publications

- AMReX-Astro Microphysics Team, Bhargava, K., Bishop, A., **Chen, Z.**, Fan, D., Fields, C., Jacobs, A. M., Johnson, E. T., Katz, M., Krumholz, M., Malone, C., Nonaka, A., Sharda, P., Smith Clark, A., Timmes, F., Wibking, B., Willcox, D. E., & Zingale, M. (2026): *AMReX-Astrophysics Microphysics: A set of microphysics routines for astrophysical simulation codes based on the AMReX library.* DOI: 10.21105/joss.10422
- Zingale, M., Bhargava, K., Brady, R., **Chen, Z.**, Guichandut, S., Johnson, E. T., Katz, M., & Smith Clark, A. (2025): *The Challenges of Modeling Astrophysical Reacting Flows.* DOI: 10.1088/1742-6596/2997/1/012007
- Zingale, M., **Chen, Z.**, Johnson, E. T., Katz, M. P., & Smith Clark, A. (2024): *Strong Coupling of Hydrodynamics and Reactions in Nuclear Statistical Equilibrium for Modeling Convection in Massive Stars.* DOI: 10.3847/1538-4357/ad8a66
- Zingale, M., **Chen, Z.**, Rasmussen, M., Polin, A., Katz, M., Smith Clark, A., & Johnson, E. T. (2024): *Sensitivity of Simulations of Double-detonation Type Ia Supernovae to Integration Methodology.* DOI: 10.3847/1538-4357/ad3441
- Clark, A., Johnson, E., **Chen, Z.**, Eiden, K., Zingale, M., Boyd, B., Johnson, P., & DaCosta, L.. (2024): *pynucastro 2.1: an update on the development of a python library for nuclear astrophysics.* DOI: 10.1088/1742-6596/2742/1/012003
- **Chen, Z.**, Johnson, E. T., Katz, M., Smith Clark, A., Boyd, B., & Zingale, M. (2024): *A Framework for Exploring Nuclear Physics Sensitivity in Numerical Simulations.* DOI: 10.1088/1742-6596/2742/1/012021
- **Chen, Z.**, Zingale, M., & Eiden, K. (2023): *Sensitivity of He Flames in X-Ray Bursts to Nuclear Physics.* DOI: 10.3847/1538-4357/acec72
- Smith, A. I., Johnson, E. T., **Chen, Z.**, Eiden, K., Willcox, D. E., Boyd, B., Cao, L., DeGrendele, C. J., & Zingale, M. (2023): *pynucastro: A Python Library for Nuclear Astrophysics.* DOI: 10.3847/1538-4357/acbaff
- **Chen, Z.**, Nomerotski, A., Slosar, A., Stankus, P., & Vintskevich, S. (2023): *Astrometry in two-photon interferometry using an Earth rotation fringe scan.* DOI: 10.1103/PhysRevD.107.023015
- Keach, M., Bellavia, S., **Chen, Z.**, Crawford, J., Dolzhenko, D., Figueroa, E., Mueninghoff, A., Nomerotski, A., Schiff, J., Simovitch, R., Slosar, A., Stankus, P., & Vintskevich, S. (2022): *Increasing baselines and precision of optical interferometers using two-photon interference effects.* DOI: 10.1117/12.2632122

Conferences and Meetings

- **Contributed Talk** at CeNAM Frontiers in Nuclear Astrophysics Meeting, 2026: *Simulating Lateral Flame Propagation in Type I X-ray Bursts*
- **Poster** Presentation at OLCF User Meeting, 2025: *Simulating X-ray Burst Flame Propagation: From Parallel Plane to Full Star*
- **Poster** Presentation at Rise_Time Meeting, 2024: *Adaptive Nuclear Statistical Equilibrium for Type Ia SN*
- **Poster** Presentation at Astronum Meeting, 2023: *Sensitivity of He Flames in X-ray Bursts to Nuclear Physics*
- **Contributed Talk** at the main CeNAM Frontiers in Nuclear Astrophysics Meeting, 2023: *Sensitivity of He Flames in X-ray Bursts to Nuclear Physics*
- SULI Internship Presentation at Brookhaven National Laboratory, Summer 2021: *Two-Photon Interferometry Simulations*
- SULI Internship Presentation at Brookhaven National Laboratory, Summer 2020: *Hydrogen Intensity Mapping Simulator: interpolation improvements and radio point source compatibility with application to GaLactic and Extragalactic All-Sky MWA Survey (GLEAM) dataset*

Honors, Grants, Awards, and Achievements

- Gerry Brown Award (2025): Awarded for excellent research and travel funding
- Lourie Summer Fellowship (2024)
- Di-Tian Prize (2023): Awarded for excellent research and travel funding
- Stony Brook University, Academic Dean's List, 2017-2021
- NYS School Academic Excellence Scholarship

Skills Summary

- Languages: Python, C++, Bash
- Tools: LaTeX, NumPy, SciPy, Sphinx
- Visualization: Matplotlib, yt
- Version Control: git, github
- Platforms: Linux
- HPC System: NESRC Perlmutter, OLCF Frontiers, ALCF Polaris